

Foundations

Data Can Deceive · Causal Roadmaps with (and without) AI

Module C

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Association versus Intervention

Association – *seeing*

- Patterns already in the data
- Treatment and outcome move together
- Purely descriptive: $P(Y \mid A)$

Intervention – *doing*

- What happens if *we act*
- Outcome under a hypothetical change
- Requires assumptions: $P(Y \mid do(A))$

An observed association can come from:

- a genuine causal effect,
- confounding (a common cause),
- selection, measurement, or other bias.

The task: move from what we see to what would happen if we *acted*, with the assumptions made explicit.

Pearl's Ladder of Causation

Three levels of causal reasoning [3]:

1. **Association (Seeing)**

What does observing A tell me about Y ? $P(Y | A)$

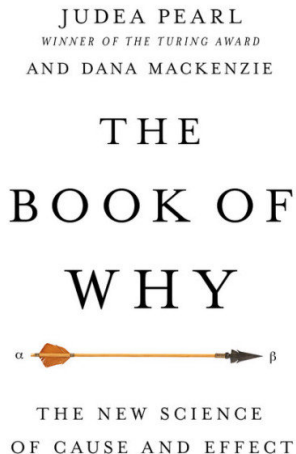
2. **Intervention (Doing)**

What happens to Y if I *do* A ? $P(Y | do(A))$

3. **Counterfactuals (Imagining)**

What *would have happened* had A been different?

Key insight: each rung needs information the rung below cannot supply. No amount of observational data alone answers an interventional question.



The Counterfactual Contrast

A causal effect compares outcomes under different interventions:

- $Y^{(1)}$: the outcome if a person *were* treated, $A = 1$
- $Y^{(0)}$: the outcome if that same person were *not*, $A = 0$

The average causal effect is $E[Y^{(1)}] - E[Y^{(0)}]$ (which may equal zero).

Formalized in Hernán & Robins' [2] *Causal Inference: What If*.

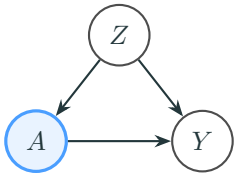
The fundamental problem of causal inference

For any one person we only ever see *one* of $Y^{(1)}, Y^{(0)}$. The other is missing, always. Causal inference is a missing-data problem we solve with design and assumptions, not with a bigger dataset.

Confounder: A Common Cause

Generic structure

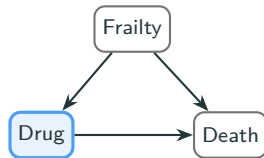
fork: $Z \rightarrow A$, $Z \rightarrow Y$



Adjust for Z to close the backdoor.

Example: frailty

sicker patients get the aggressive drug

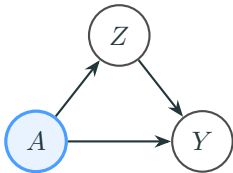


Frailty drives both treatment *and* death, so the crude contrast makes the drug look harmful.

Mediator: On the Causal Path

Generic structure

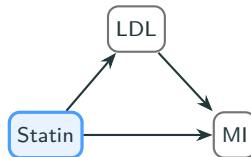
chain: $A \rightarrow Z \rightarrow Y$



Do not adjust (for a total effect).

Example: statin and LDL

the drug acts *through* lowering LDL

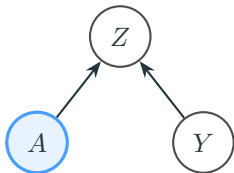


Adjust for LDL and the statin looks inert – you have removed the pathway it acts through.

Collider: A Common Effect

Generic structure

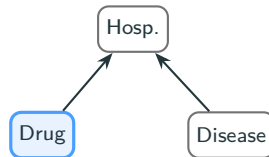
collision: $A \rightarrow Z \leftarrow Y$



Do not adjust – it opens a fake path.

Example: hospitalization

drug and disease both cause admission



Restrict to the hospitalized and drug and disease look linked.

Why “Adjust for Everything” Fails

A single reflex applied to all three does two kinds of damage:

Adjust a mediator

You block part of the effect you are trying to measure. Condition on LDL and the statin looks inert – you have thrown away the mechanism.

Adjust a collider

You *open* a path that was closed. Restrict to hospitalized patients and drug and disease look linked when in the population they are not.

The lesson: the covariate list is a set of *claims about structure*, not a bigger-is-safer knob. What to adjust for is decided by the causal question and the diagram, before the data.

Which is exactly what Module D’s roadmap makes systematic.

Design First: The Target Trial

Specify the randomized trial you *wish* you could run [1], then emulate it with the data you have:

- eligibility, treatment strategies, assignment,
- follow-up, outcome, and the causal contrast.

Time zero: align eligibility, treatment, and follow-up at one instant

Every patient's clock starts when they meet eligibility and treatment is assigned. Getting this wrong is the classic source of **immortal-time bias** (survival smuggled into the treated group) and **prevalent-user bias** (survivors of early treatment overrepresented).

Why it matters here: a clean time zero is a *design* fix for problems no downstream model can undo.

Why Vaccine Results Can Appear to Reverse

In an observational comparison, vaccinated older adults can appear to have **higher pneumonia risk** than unvaccinated adults.

A causal explanation is not the only explanation:

- Frailer, higher-risk patients may be more likely to receive the vaccine.
- Frailty also raises pneumonia risk.
- Frailty is therefore a **confounder**: a common cause of vaccination and pneumonia.

Separate real example: CAPITA

CAPITA was a randomized trial in adults aged 65 years or older, not the synthetic workshop cohort used in Modules D and F. It estimated 45.6% efficacy against a first episode of vaccine-type community-acquired pneumonia [4].

Before the roadmap, test two design habits:
classify causal roles and align time zero.

Poll 1 – Classify the Variable

Among **hospitalized** patients only, a drug's side effects and the underlying disease both raise the chance of admission. You analyze within the hospitalized group.

Hospitalization (the variable you restricted on) is a:

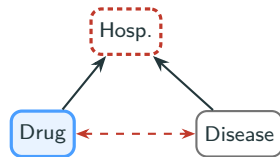
- A. Confounder
- B. Mediator
- C. Collider

Poll 1 – Solution

C – Collider.

Side effects and disease both cause admission, so both arrows point *into* hospitalization.

Restricting to it opens a fake link.



dashed = induced association

Poll 2 – What Is Wrong With This Comparison?

Patients who filled a prescription for the drug *within the first year* are labelled “treated”; everyone else is “untreated.” Survival is compared from cohort entry.

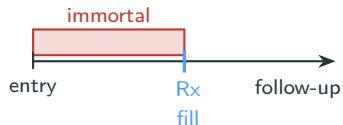
The main problem is:

- A. Too small a sample
- B. Immortal time – the treated had to survive long enough to fill it
- C. Measurement error in the outcome

B – Immortal time.

To be labelled “treated,” a patient must survive to the fill date. That guaranteed window biases toward the drug.

Fix: start time zero at treatment assignment.



survival in the shaded window is guaranteed

Poll 3 – Which Is Safe to Adjust For?

In a statin study you could adjust for LDL cholesterol measured *at baseline*, or LDL measured *at 6 months on treatment*. You want the **total** effect of the statin on heart attacks.

Which is safe to adjust for?

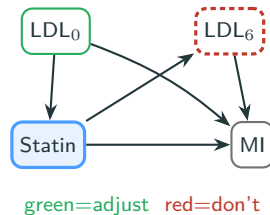
- A. Baseline LDL
- B. On-treatment LDL
- C. Both, to be safe

Poll 3 – Solution

A – baseline LDL only.

Baseline LDL is a pre-treatment confounder (adjust). On-treatment LDL is a mediator (do not).

Same variable, two roles – set by timing.



Poll 4 – No Comparison Group

Clinical guidelines mandate the drug for *everyone* with severe disease, so in your data there are **zero untreated** severe patients.

For the severe subgroup, you:

- A. Can still estimate the effect by adjusting
- B. Cannot estimate it – there is no comparison group
- C. Should impute the missing untreated patients

Poll 4 – Solution

B – positivity violation.

Identification needs both arms within each stratum. With no untreated severe patients, that effect is not in the data.

Imputing the arm is extrapolation. Module D will formalize this as a positivity and overlap check.

	Treated	Untreated
mild	• • •	• •
severe	• • •	(none)

no untreated severe patients = no contrast

Module D turns these foundations into a reusable roadmap.

Carry forward the target trial, time zero, DAG roles, and positivity.

References

- [1] Hernán MA, Robins JM. Using big data to emulate a target trial when a randomized trial is not available. *American Journal of Epidemiology*, 183(8):758–764, 2016.
- [2] Hernán MA, Robins JM. *Causal Inference: What If*. Chapman & Hall/CRC, 2020.
- [3] Pearl J, Mackenzie D. *The Book of Why: The New Science of Cause and Effect*. Basic Books, 2018.
- [4] Bonten MJM, Huijts SM, Bolkenbaas M, et al. Polysaccharide conjugate vaccine against pneumococcal pneumonia in adults. *New England Journal of Medicine*, 372(12):1114–1125, 2015.